

## Results

Candidate Outcome

## Candidate Outcome I

The generated loop selected exp-mlp-tanh-64 (One-hidden-layer tanh MLP, MLP) after evaluating 4 candidate(s) from a proposed set of 5. The nearest\_centroid\_baseline (nearest-centroid) baseline reached 82.6% test\_accuracy, while the selected candidate reached 89.4%, an absolute change of 6.8%. The selected model has 50890 parameters. The transformer-candidate evaluated flag is true, and the candidate budget exhausted flag is true, which means the ledger records 1 deferred proposal(s) rather than expanding the run automatically.

## Candidate Outcome I

The benchmark score is 1. That score is not a model-quality claim by itself; it is a compact grading artifact for the methods contract: metric improvement, budget compliance, offline execution, and selected-candidate recording. Rank-stability diagnostics report that the selected candidate is top ranked in 72.5% of deterministic bootstrap resamples, with runner-up `exp-mlp-relu-32`. The candidate score figure is registered as fig. 1, the confusion matrix as fig. 3, the per-class diagnostic as fig. 4, the learning curves as fig. 5, the complexity diagnostic as fig. 7, the selected-candidate error examples as fig. 8, the candidate lifecycle diagnostic as fig. 9, rank

## Candidate Outcome I

stability as fig. 2, the training-dynamics diagnostic as fig. 6, the final readiness matrix as fig. 20, and the process closure as fig. 21.

## Candidate Outcome I

Table 1: Candidate accuracy intervals from `output/data/ml_candidate_intervals.json`; intervals describe the fixed local test split.

| Candidate            | Status   | Correct/test | Accuracy | Wilson 95% interval |
|----------------------|----------|--------------|----------|---------------------|
| baseline             | baseline | 413/500      | 82.6%    | 79.0% to 85.7%      |
| softmax linear       | rejected | 441/500      | 88.2%    | 85.1% to 90.7%      |
| mlp relu 32          | rejected | 443/500      | 88.6%    | 85.5% to 91.1%      |
| tiny patch attention | rejected | 152/500      | 30.4%    | 26.5% to 34.6%      |
| mlp tanh 64          | accepted | 447/500      | 89.4%    | 86.4% to 91.8%      |

## Candidate Outcome II

Table 2: Candidate rank-stability table from output/data/ml\_candidate\_rank\_stability.json; frequencies are deterministic local bootstrap summaries.

| Candidate            | Observed rank | Top-rank frequency | Mean rank | Accuracy |
|----------------------|---------------|--------------------|-----------|----------|
| softmax linear       | 3             | 4.2%               | 2.554     | 88.2%    |
| mlp relu 32          | 2             | 23.3%              | 2.130     | 88.6%    |
| tiny patch attention | 4             | 0.0%               | 4.000     | 30.4%    |
| mlp tanh 64          | 1             | 72.5%              | 1.316     | 89.4%    |

## Run-Derived Figures

# Run-Derived Figures I

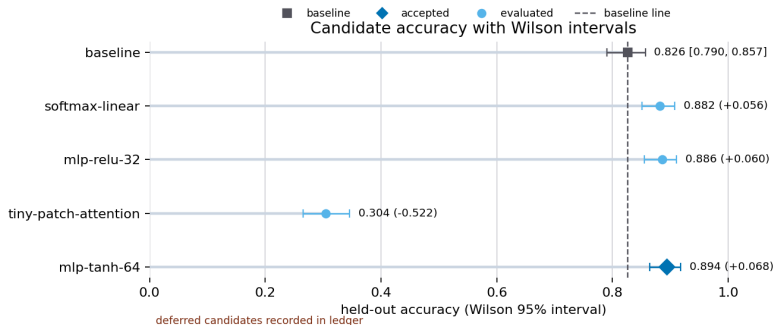


Figure 1: Held-out accuracy with Wilson intervals for the baseline and evaluated candidates from output/data/ml\_candidate\_intervals.json; accepted candidate exp-mlp-tanh-64 improves accuracy from 82.6% to 89.4% on the fixed subset, with deferred proposals kept in the candidate ledger. Generation method: Lollipop accuracy comparison with Wilson interval error bars and direct labels. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Run-Derived Figures II

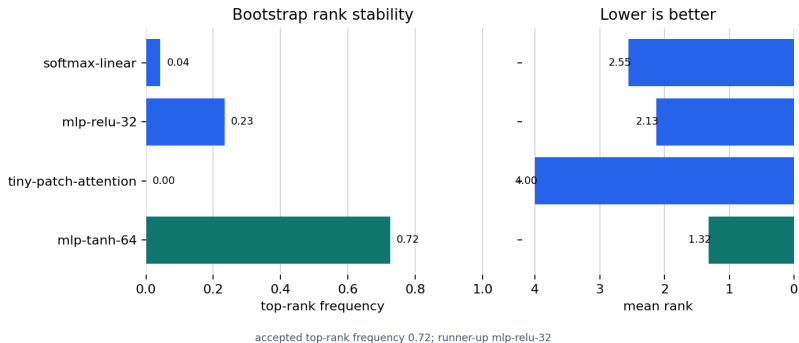
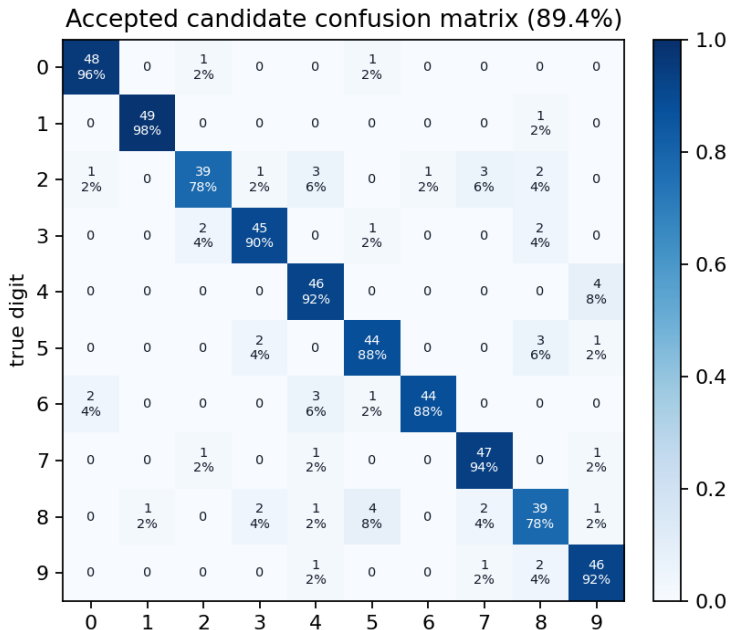


Figure 2: Rank-stability summary for exp-mlp-tanh-64 from output/data/ml\_candidate\_rank\_stability.json; deterministic local bootstrap resampling shows how often each evaluated candidate ranks first under the fixed test split. Generation method: Bootstrap top-rank frequency and mean-rank comparison. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Run-Derived Figures III



## Training And Error Diagnostics

# Training And Error Diagnostics I

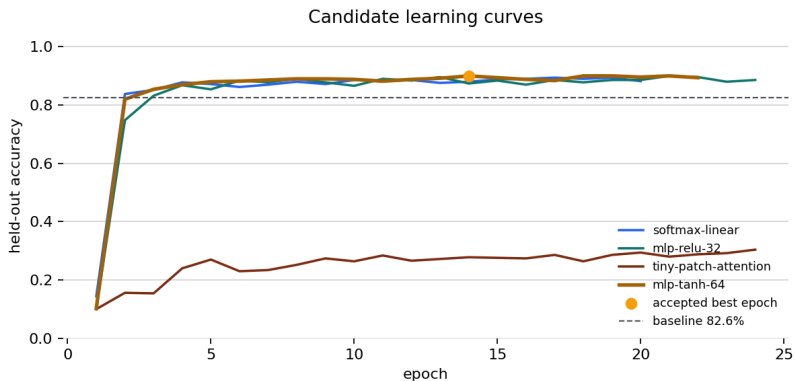


Figure 5: Epoch-level held-out accuracy curves for evaluated candidates from output/data/ml\_training\_history.csv; the accepted curve is visually emphasized for exp-mlp-tanh-64. Generation method: Epoch-level held-out accuracy lines with accepted best-epoch marker. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Training And Error Diagnostics II

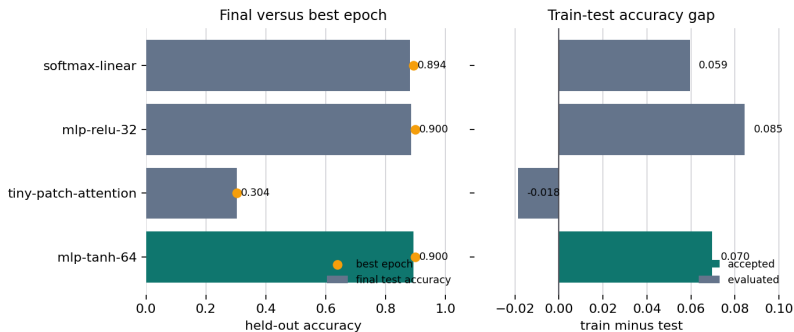


Figure 6: Configured-training dynamics for evaluated candidates from output/data/ml\_training\_diagnostics.json; exp-mlp-tanh-64 is highlighted while best-epoch markers and train-test gaps remain bounded to the local run. Generation method: Final and best-epoch accuracy bars plus train-test gap bars. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Training And Error Diagnostics III

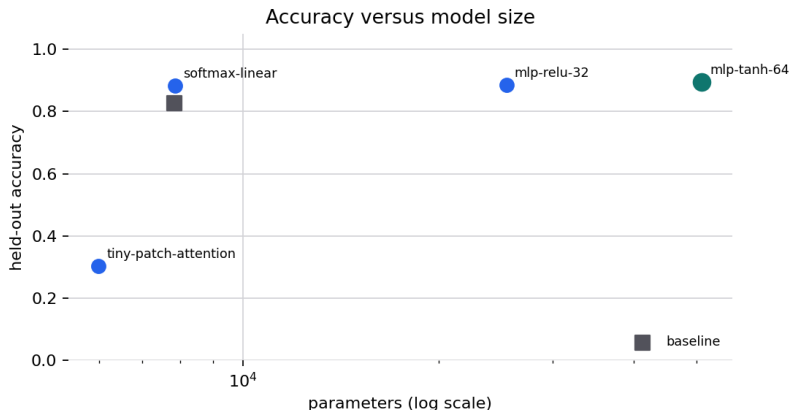


Figure 7: Parameter-count versus held-out accuracy for the baseline and evaluated candidates from `output/data/ml_task_results.json`; the accepted candidate is highlighted without claiming a general scaling law. Generation method: Log-parameter scatter plot against held-out accuracy. Registry metadata records the generation method, source artifact, and claim boundary for validation.

# Training And Error Diagnostics IV

## Accepted candidate error examples

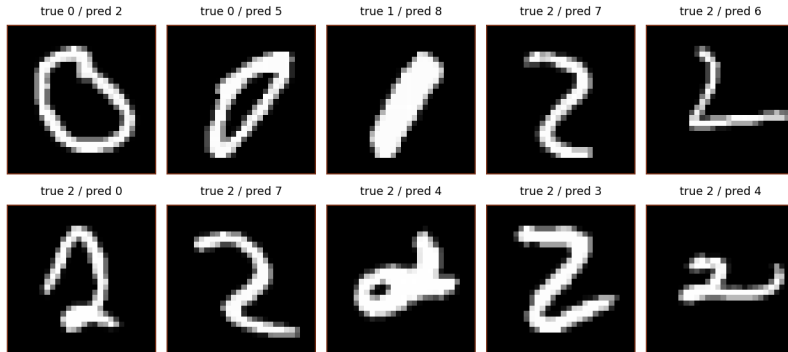


Figure 8: First accepted-candidate error examples for exp-mlp-tanh-64, sourced from output/data/ml\_error\_examples.json and data/mnist\_small.npz; these images support qualitative diagnosis only. Generation method: Deterministic grid of the first accepted-candidate misclassifications. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Training And Error Diagnostics V

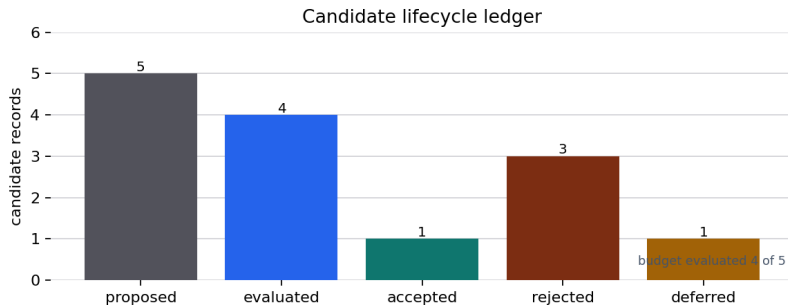


Figure 9: Candidate lifecycle ledger from output/data/ml\_candidate\_ledger.json: 4 evaluated out of 5 proposed candidates, with deferred proposals kept visible instead of executed automatically. Generation method: Candidate lifecycle status-count bar chart. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Training And Error Diagnostics I

The selected candidate's best held-out epoch is 14, its final learning rate is 0.144, its train-loss reduction is 4.161, and its final train-test accuracy gap is 7.0%. These values come from the configured training diagnostics rather than from a manually maintained result summary.

# Training And Error Diagnostics I

Table 3: Configured-training diagnostics from output/data/ml\_training\_diagnostics.json.

| Candidate   | Status   | Best epoch | Best test accuracy | Final test accuracy | Train-test gap | Loss reduction | Final learning rate |
|-------------|----------|------------|--------------------|---------------------|----------------|----------------|---------------------|
| softmax     | rejected | 17         | 89.4%              | 88.2%               | 5.9%           | 2.917          | 0.273               |
| linear      |          |            |                    |                     |                |                |                     |
| mlp relu 32 | rejected | 21         | 90.0%              | 88.6%               | 8.5%           | 4.341          | 0.178               |
| tiny patch  | rejected | 24         | 30.4%              | 30.4%               | -1.8%          | 0.165          | 0.214               |
| attention   |          |            |                    |                     |                |                |                     |
| mlp tanh 64 | accepted | 14         | 90.0%              | 89.4%               | 7.0%           | 4.161          | 0.144               |

## Diagnostic Analysis

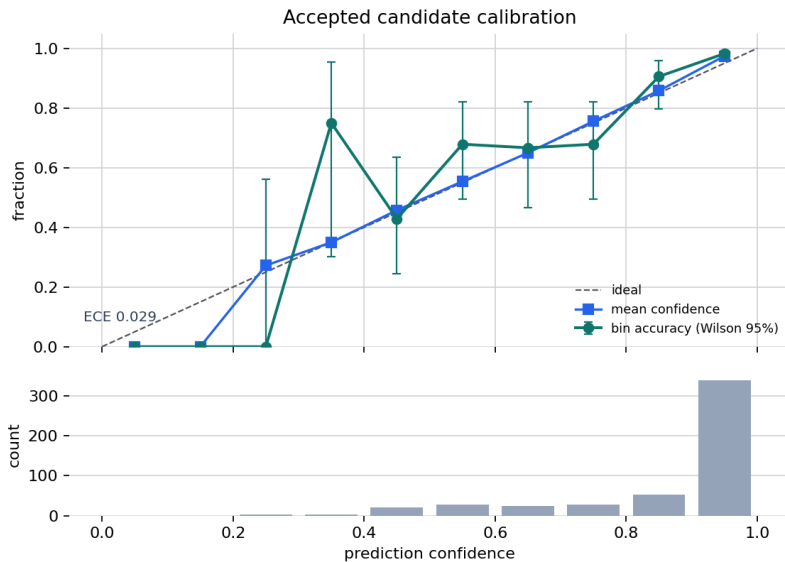
## Diagnostic Analysis I

The selected candidate macro F1 is 89.4%, with a held-out accuracy interval of 86.4% to 91.8%. Probability diagnostics report expected calibration error 2.9% and 11 high-confidence error(s). The top non-diagonal confusion pair is 4  $\rightarrow$  9 (4), and the minimum selected-candidate accuracy across deterministic robustness transforms is 80.0%. These values are descriptive diagnostics for the validated local run, not external benchmark claims.

## Classification And Error Structure I

The class-level and error-pattern diagnostics are intentionally visual-first: fig. 10 checks confidence calibration, fig. 11 separates precision, recall, and F1 by digit, and fig. 12 ranks the non-diagonal confusions that most shape the selected-candidate error profile. The accompanying tables preserve the same values for audit and downstream comparison.

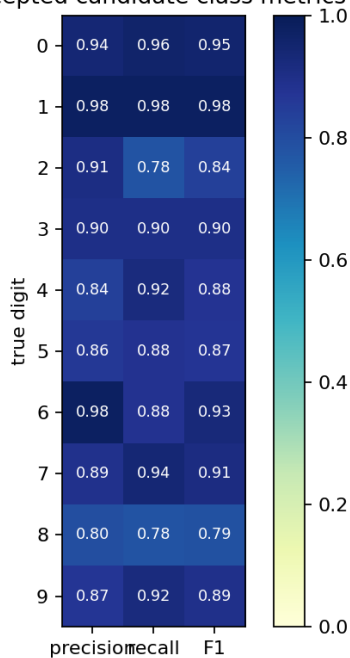
# Classification And Error Structure I



# Classification And Error Structure II

Figure 10: Reliability curve for exp-mlp-tanh-64 from output/data/ml\_calibration\_report.json; expected calibration error and bin counts summarize the accepted candidate on the fixed local test split. Generation method: Reliability curve with confidence-bin support histogram. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Accepted candidate class metrics



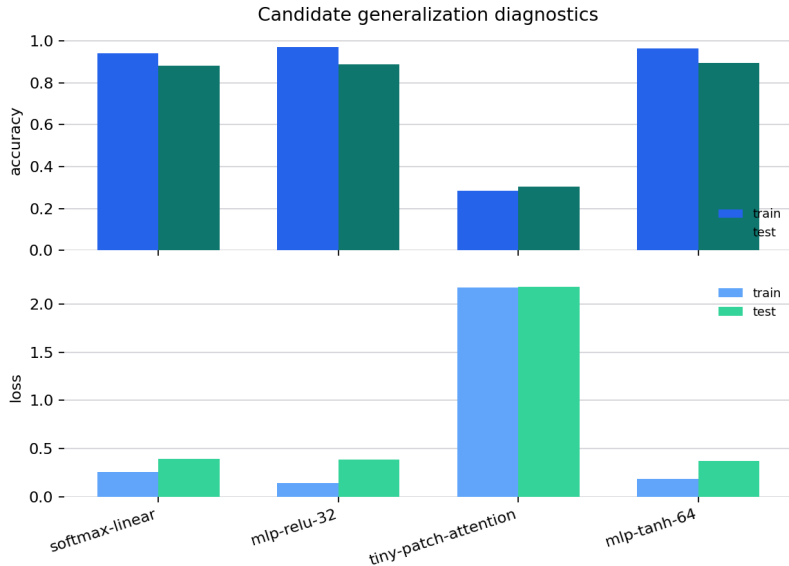
## Uncertainty, Generalization, And Perturbation I

Additional uncertainty and matched-comparison diagnostics report bootstrap accuracy interval 86.4% to 92.0%, bootstrap macro-F1 interval 86.5% to 91.9%, paired net accuracy gain 6.8%, exact McNemar p-value 0.000, mean correct-prediction confidence 90.9%, mean error confidence 63.1%, and 22 low-margin selected-candidate prediction(s). These diagnostics are generated from the fixed local test split and are reported as run evidence, not as population-level certification.

## Uncertainty, Generalization, And Perturbation I

fig. 13 compares train and test behavior, fig. 14 shows deterministic no-retrain perturbation results, fig. 15 summarizes confidence and margin distributions, fig. 16 shows local resampling intervals, and fig. 17 exposes the matched selected-versus-baseline correctness table used by the paired comparison.

# Uncertainty, Generalization, And Perturbation I



## Uncertainty, Generalization, And Perturbation II

Figure 13: Train/test accuracy and loss for evaluated candidates from `output/data/ml_classification_diagnostics.json`; the plot exposes local generalization gaps without claiming full-dataset behavior. Generation method: Grouped train/test accuracy and loss bars by evaluated candidate. Registry metadata records the generation method, source artifact, and claim boundary for validation.

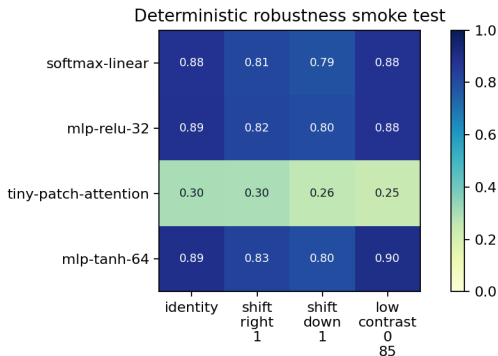
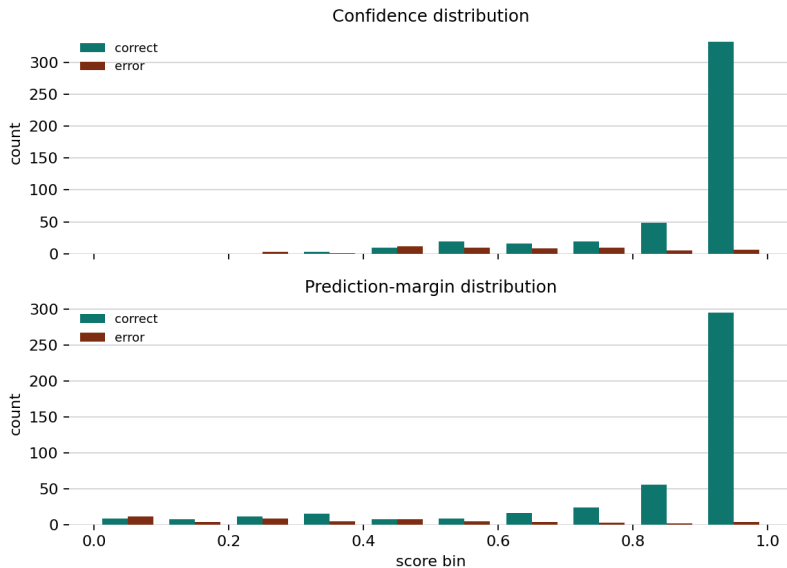


Figure 14: Accuracy for evaluated candidates under identity, one-pixel shifts, and low contrast from `output/data/ml_robustness_report.json`; the deterministic transforms provide a bounded smoke test only. Generation method: Candidate-by-transform accuracy heatmap for deterministic perturbations. Registry metadata records the generation method, source artifact, and claim boundary for validation.

# Uncertainty, Generalization, And Perturbation III



## Uncertainty, Generalization, And Perturbation IV

Figure 15: Confidence and prediction-margin histograms for exp-mlp-tanh-64 from output/data/ml\_probability\_diagnostics.json; the figure separates correct and incorrect local test predictions. Generation method: Confidence and margin histograms split by correctness. Registry metadata records the generation method, source artifact, and claim boundary for validation.

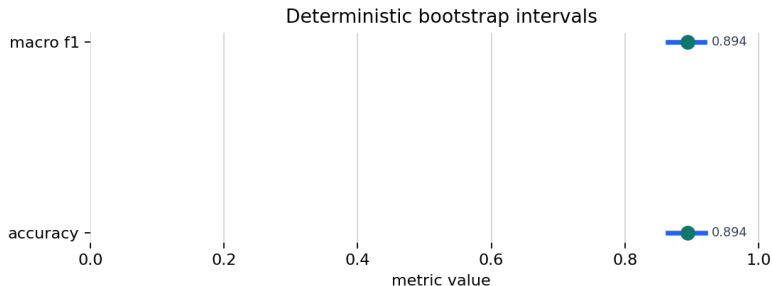
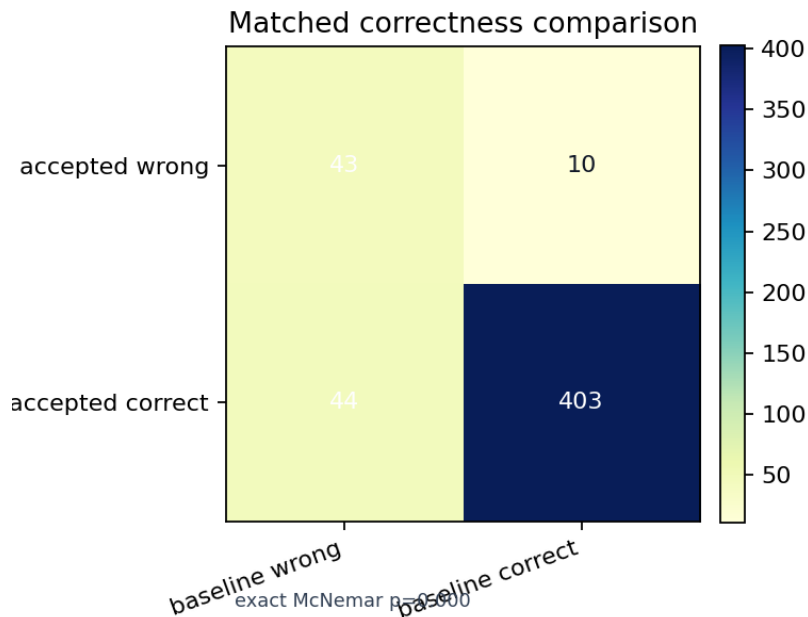


Figure 16: Deterministic percentile-bootstrap intervals for exp-mlp-tanh-64 from output/data/ml\_bootstrap\_intervals.json; the intervals summarize local sampling variation for accuracy and macro F1. Generation method: Horizontal percentile-bootstrap interval plot. Registry metadata records the generation method, source artifact, and claim boundary for validation.



## Probability Quality And Selective Prediction I

Probability-quality diagnostics report Brier score 0.161, negative log likelihood 0.361, top-2 accuracy 95.6%, and Cohen kappa 0.882 for the selected candidate. At the highest configured confidence threshold, retained coverage is 67.8% and selective accuracy is 98.2%.

The selective-prediction view in fig. 18 reports the configured confidence-threshold trade-off: retaining fewer predictions can raise selective accuracy, but the coverage table keeps that trade-off explicit. The candidate probability-quality comparison in fig. 19 keeps accuracy separate from proper-score behavior, so the selected candidate is not treated as automatically best on every diagnostic axis.

# Probability Quality And Selective Prediction I

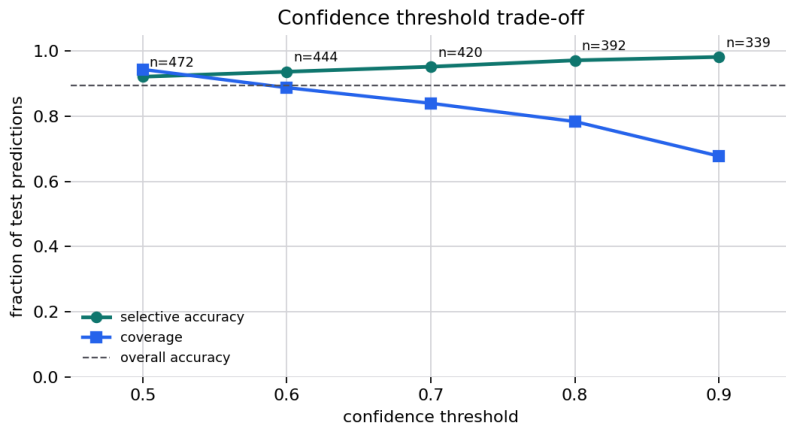


Figure 18: Confidence-threshold trade-off for exp-mlp-tanh-64 from output/data/ml\_statistical\_summary.json; the plot compares retained coverage, selective accuracy, and the unthresholded accepted-candidate accuracy on the fixed local split. Generation method: Confidence-threshold coverage and selective-accuracy line chart. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Probability Quality And Selective Prediction II

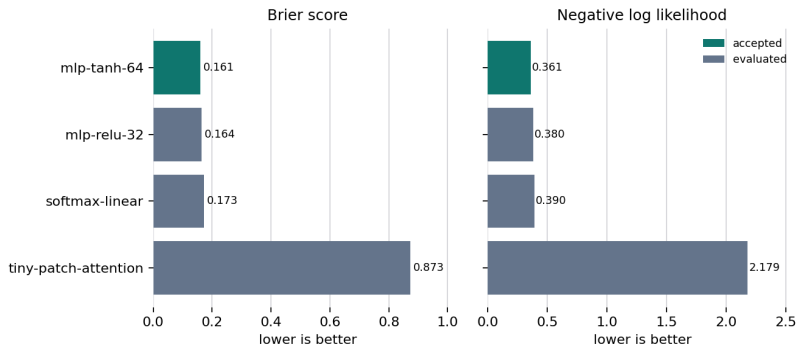


Figure 19: Brier score and negative log likelihood for evaluated candidates from `output/data/ml_statistical_summary.json`; lower values indicate better probability quality within the configured local run, and the accepted candidate is highlighted. Generation method: Brier score and negative-log-likelihood bar comparison. Registry metadata records the generation method, source artifact, and claim boundary for validation.

## Probability Quality And Selective Prediction III

Table 12: Accepted-candidate statistical summary from output/data/ml\_statistical\_summary.json.

| Statistic                  | Value |
|----------------------------|-------|
| Accuracy                   | 89.4% |
| Balanced accuracy          | 89.4% |
| Macro F1                   | 89.4% |
| Top-2 accuracy             | 95.6% |
| Cohen kappa                | 0.882 |
| Brier score                | 0.161 |
| Negative log likelihood    | 0.361 |
| Expected calibration error | 2.9%  |

## Probability Quality And Selective Prediction IV

Table 13: Selective-accuracy threshold table from output/data/ml\_statistical\_summary.json.

| Confidence threshold | Coverage | Selective accuracy | Retained | Errors |
|----------------------|----------|--------------------|----------|--------|
| 50.0%                | 94.4%    | 92.2%              | 472      | 37     |
| 60.0%                | 88.8%    | 93.7%              | 444      | 28     |
| 70.0%                | 84.0%    | 95.2%              | 420      | 20     |
| 80.0%                | 78.4%    | 97.2%              | 392      | 11     |
| 90.0%                | 67.8%    | 98.2%              | 339      | 6      |

# Probability Quality And Selective Prediction V

Table 14: Candidate probability-quality table from output/data/ml\_statistical\_summary.json.

| Candidate      | Accuracy | Top-2 accuracy | Brier | NLL   | Mean confidence |
|----------------|----------|----------------|-------|-------|-----------------|
| softmax linear | 88.2%    | 95.8%          | 0.173 | 0.390 | 85.1%           |
| mlp relu 32    | 88.6%    | 95.6%          | 0.164 | 0.380 | 89.9%           |
| tiny patch     | 30.4%    | 46.6%          | 0.873 | 2.179 | 13.1%           |
| attention      |          |                |       |       |                 |
| mlp tanh 64    | 89.4%    | 95.6%          | 0.161 | 0.361 | 87.9%           |

## Candidate Ledger

# Candidate Ledger I

Table 15: Candidate lifecycle ledger from `output/data/ml_candidate_ledger.json`.

| Candidate            | Model                | Status   | Test accuracy | Parameters |
|----------------------|----------------------|----------|---------------|------------|
| baseline             | nearest-centroid     | baseline | 82.6%         | 7840       |
| softmax linear       | softmax regression   | rejected | 88.2%         | 7850       |
| mlp relu 32          | MLP                  | rejected | 88.6%         | 25450      |
| tiny patch attention | tiny patch-attention | rejected | 30.4%         | 5994       |
| mlp tanh 64          | MLP                  | accepted | 89.4%         | 50890      |
| mlp relu 64 deferred | MLP                  | deferred | N/A           | 0          |

## Candidate Ledger I

The candidate-selection audit separates the objective ranking from descriptive diagnostics. It records the configured metric, Wilson interval, probability quality, parameter count, and deterministic tie-break context for each evaluated candidate. The diagnostic-boundary table states what each generated surface supports and what it does not support.

## Candidate Ledger I

Table 16: Candidate-selection audit from `output/data/ml_candidate_selection_audit.json`; the objective metric ranks candidates, while probability diagnostics and parameter counts audit the chosen tie-break context.

| Rank | Candidate            | Status   | Accuracy | Wilson 95%     | Brier | NLL   | Parameters |
|------|----------------------|----------|----------|----------------|-------|-------|------------|
| 1    | mlp tanh 64          | accepted | 89.4%    | 86.4% to 91.8% | 0.161 | 0.361 | 50890      |
| 2    | mlp relu 32          | rejected | 88.6%    | 85.5% to 91.1% | 0.164 | 0.380 | 25450      |
| 3    | softmax linear       | rejected | 88.2%    | 85.1% to 90.7% | 0.173 | 0.390 | 7850       |
| 4    | tiny patch attention | rejected | 30.4%    | 26.5% to 34.6% | 0.873 | 2.179 | 5994       |

# Candidate Ledger I

Table 17: Diagnostic claim-boundary table from `output/data/ml_diagnostic_boundary.json`.

| Surface                 | Source                | Method                                                                           | Supports                                                  | Does not support                                                                 |
|-------------------------|-----------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------|
| objective selection     | task results          | rank evaluated candidates by configured held-out metric and deterministic tie... | accepted-candidate selection within this fixed local task | full MNIST state-of-the-art, external benchmark leadership, or universal mode... |
| descriptive diagnostics | class diagnostics     | per-class metrics, calibration, probability quality, and paired comparison       | local error analysis and uncertainty description          | population-level certification or deployment readiness                           |
| robustness smoke test   | robustness report     | deterministic no-retrain transforms applied to the fixed test split              | small perturbation smoke-test evidence                    | adversarial robustness or distribution-shift robustness                          |
| artifact integrity      | integrity attestation | local SHA-256 checks over declared inputs and generated artifacts                | local artifact integrity evidence for the run             | external signing, production SLSA compliance, or runtime intrusion detection     |
| review governance       | review decisions      | deferred generated review decisions with human review required                   | readiness for human review                                | machine self-approval or publication acceptance                                  |

## Readiness And Review Artifacts

## Readiness And Review Artifacts I

The broader AutoResearch run writes the reproducibility, benchmark, review, and manuscript-hydration surfaces summarized below. The schema manifest records 31 schema-versioned governance payload(s), and the local research-object manifest records 84 observed artifact record(s) with checksums and approval state false. The phase ledger records 3 settlement pass(es), while the figure-quality report covers 25 registered figure(s) with validity false.

## Readiness And Review Artifacts I

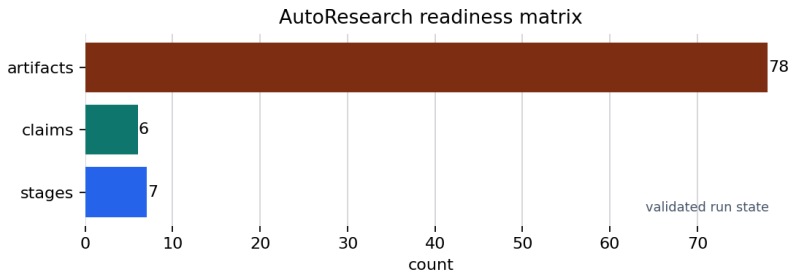


Figure 20: Validated AutoResearch run with 7 stages, 6 supported claims, and 78 required artifacts from output/data/autoresearch\_loop.json; the count summarizes readiness artifacts, not human approval. Generation method: Horizontal count summary from final loop metrics. Registry metadata records the generation method, source artifact, and claim boundary for validation.

# Readiness And Review Artifacts II

File-backed AutoResearch closure



Figure 21: File-backed AutoResearch closure from program through review, with 6 supported claims and readiness passed; review remains a deferred human decision and the provenance path remains inspectable. Generation method: File-backed process-flow diagram from final loop state. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Table 18: Reviewed upstream analysis and evidence paths from `output/reports/artifact_manifest.json`; unreviewed paths are omitted, and that terminal manifest remains the complete byte authority.

| Artifact                                             | Role                    |
|------------------------------------------------------|-------------------------|
| output/data/autoresearch_claims.json                 | Loop artifact           |
| output/data/autoresearch_evidence_overview.json      | Evidence registry       |
| output/data/autoresearch_integrity_attestation.json  | Security evidence       |
| output/data/autoresearch_inventory_export.json       | Security evidence       |
| output/data/autoresearch_loop.json                   | Loop artifact           |
| output/data/autoresearch_phase_ledger.json           | Run or candidate ledger |
| output/data/autoresearch_plan.json                   | Loop artifact           |
| output/data/autoresearch_review_packet.json          | Review packet           |
| output/data/autoresearch_schema_manifest.json        | Loop artifact           |
| output/data/autoresearch_security_profile.json       | Security evidence       |
| output/data/autoresearch_stage_matrix.csv            | Loop artifact           |
| output/data/autoresearch_supply_chain_inventory.json | Security evidence       |
| output/data/autoresearch_threat_model.json           | Security evidence       |
| output/data/benchmark_boundary.json                  | Benchmark grading       |

## Readiness And Review Artifacts III

|                                                         |                         |
|---------------------------------------------------------|-------------------------|
| output/data/benchmark_scores.json                       | Benchmark grading       |
| output/data/figure_quality_report.json                  | Loop artifact           |
| output/data/figure_style.json                           | Loop artifact           |
| output/data/idea_ledger.json                            | Run or candidate ledger |
| output/data/ml_bootstrap_intervals.json                 | Loop artifact           |
| output/data/ml_calibration_bin_intervals.json           | Loop artifact           |
| output/data/ml_calibration_report.json                  | Loop artifact           |
| output/data/ml_candidate_intervals.json                 | Loop artifact           |
| output/data/ml_candidate_ledger.json                    | Run or candidate ledger |
| output/data/ml_candidate_rank_stability.json            | Loop artifact           |
| output/data/ml_candidate_selection_audit.json           | Loop artifact           |
| output/data/ml_class_balance.json                       | Loop artifact           |
| output/data/ml_classification_diagnostics.json          | Loop artifact           |
| output/data/ml_confusion_matrix.csv                     | Loop artifact           |
| output/data/ml_diagnostic_boundary.json                 | Loop artifact           |
| output/data/ml_error_examples.json                      | Loop artifact           |
| output/data/ml_paired_comparison.json                   | Loop artifact           |
| output/data/ml_prediction_records.json                  | Loop artifact           |
| output/data/ml_probability_diagnostics.json             | Loop artifact           |
| output/data/ml_robustness_report.json                   | Loop artifact           |
| output/data/ml_statistical_summary.json                 | Loop artifact           |
| output/data/ml_task_results.json                        | Loop artifact           |
| output/data/ml_training_diagnostics.json                | Loop artifact           |
| output/data/ml_training_history.csv                     | Loop artifact           |
| output/data/mnist_task_config.json                      | Loop artifact           |
| output/data/research_object_manifest.json               | Loop artifact           |
| output/data/research_program.json                       | Loop artifact           |
| output/data/review_decisions.json                       | Review packet           |
| output/data/run_ledger.json                             | Run or candidate ledger |
| output/figures/autoresearch_candidate_lifecycle.png     | Generated figure        |
| output/figures/autoresearch_closure_flow.png            | Generated figure        |
| output/figures/autoresearch_integrity_chain.png         | Generated figure        |
| output/figures/autoresearch_security_control_matrix.png | Generated figure        |
| output/figures/autoresearch_stage_matrix.png            | Generated figure        |
| output/figures/figure_registry.json                     | Loop artifact           |
| output/figures/ml_bootstrap_intervals.png               | Generated figure        |
| output/figures/ml_calibration_reliability.png           | Generated figure        |

## Readiness And Review Artifacts IV

|                                                       |                   |
|-------------------------------------------------------|-------------------|
| output/figures/ml_candidate_rank_stability.png        | Generated figure  |
| output/figures/ml_candidate_scores.png                | Generated figure  |
| output/figures/ml_classification_metrics_heatmap.png  | Generated figure  |
| output/figures/ml_complexity_accuracy.png             | Generated figure  |
| output/figures/ml_confusion_matrix.png                | Generated figure  |
| output/figures/ml_confusion_pairs.png                 | Generated figure  |
| output/figures/ml_generalization_gap.png              | Generated figure  |
| output/figures/ml_learning_curves.png                 | Generated figure  |
| output/figures/ml_paired_correctness.png              | Generated figure  |
| output/figures/ml_per_class_accuracy.png              | Generated figure  |
| output/figures/ml_probability_margin_distribution.png | Generated figure  |
| output/figures/ml_probability_quality.png             | Generated figure  |
| output/figures/ml_robustness_matrix.png               | Generated figure  |
| output/figures/ml_selective_accuracy.png              | Generated figure  |
| output/figures/ml_training_dynamics.png               | Generated figure  |
| output/figures/mnist_class_balance.png                | Generated figure  |
| output/figures/mnist_error_examples.png               | Generated figure  |
| output/figures/mnist_subset_contact_sheet.png         | Generated figure  |
| output/reports/autoresearch_evidence_overview.md      | Evidence registry |
| output/reports/autoresearch_loop.json                 | Loop artifact     |
| output/reports/autoresearch_loop.md                   | Loop artifact     |
| output/reports/autoresearch_review_packet.md          | Review packet     |
| output/reports/autoresearch_security_review.md        | Review packet     |
| output/reports/autoresearch_summary.md                | Loop artifact     |
| output/reports/benchmark_readiness_smoke.json         | Benchmark grading |
| output/reports/ml_benchmark_score.json                | Benchmark grading |
| output/reports/ml_experiment_report.md                | Loop artifact     |
| output/reports/test_results.json                      | Loop artifact     |
| output/reports/test_results.md                        | Loop artifact     |

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## Readiness And Review Artifacts V

Table 19: Review-gate decisions from `output/data/review_decisions.json`.

| Gate            | Required | Decision | Rationale                                                                                               |
|-----------------|----------|----------|---------------------------------------------------------------------------------------------------------|
| proposal_review | True     | deferred | Decision is read from <code>human_review.yaml</code> when present; generated readiness is not approval. |
| evidence_review | True     | deferred | Decision is read from <code>human_review.yaml</code> when present; generated readiness is not approval. |

Table 20: Benchmark grading table from output/data/benchmark\_scores.json.

| Benchmark task  | Status | Score | Grading output                      |
|-----------------|--------|-------|-------------------------------------|
| readiness-smoke | graded | 1     | output/reports/benchmark_readiness_ |
| ml-loop-score   | graded | 1     | output/reports/ml_benchmark_score.j |

Table 21: Deterministic phase ledger from output/data/autoresearch\_phase\_ledger.json; settlement order is not an autonomy claim.

| Phase                    | Order | Group      | Observed artifacts | Description                                             |
|--------------------------|-------|------------|--------------------|---------------------------------------------------------|
| intrinsic readiness      | 1     | readiness  | 3                  | validate configured project-intrinsic contracts         |
| core artifacts           | 2     | loop       | 0                  | write plan, stage matrix, and provisional loop outputs  |
| evidence registry        | 3     | evidence   | 2                  | write local evidence registry                           |
| ml task                  | 4     | ml         | 54                 | run fixed-seed bounded candidate evaluation             |
| method contract          | 5     | governance | 3                  | write program, idea, run, review, and benchmark ledgers |
| provisional payloads     | 6     | settlement | 12                 | refresh loop payloads before extrinsic validation       |
| security artifacts       | 7     | security   | 10                 | write local security and integrity evidence             |
| final visuals            | 8     | figures    | 54                 | write final registry-backed figures                     |
| manuscript hydration     | 9     | manuscript | 9                  | write variables, provenance, and figure blocks          |
| readiness manifest       | 10    | settlement | 12                 | refresh checksum manifest before extrinsic validation   |
| schema manifest          | 11    | schema     | 1                  | write generated JSON schema-version manifest            |
| research object manifest | 12    | packaging  | 1                  | write local research-object manifest                    |

## Readiness And Review Artifacts VIII

|                                |    |            |    |                                                      |
|--------------------------------|----|------------|----|------------------------------------------------------|
| extrinsic readiness            | 13 | readiness  | 3  | validate generated artifacts and extrinsic contracts |
| final schema manifest          | 14 | schema     | 1  | refresh schema manifest after final payload updates  |
| final research object manifest | 15 | packaging  | 1  | refresh local research-object manifest               |
| artifact manifest              | 16 | settlement | 12 | write final artifact checksum manifest               |

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## Readiness And Review Artifacts IX

Table 22: Figure-quality checks from output/data/figure\_quality\_report.json; 25 registered figure(s) were checked.

| Figure                                 | Pixels   | Variance | Source exists | Nonblank |
|----------------------------------------|----------|----------|---------------|----------|
| fig:autoresearch_candidate_cycle       | 1184x480 | 0.082    | True          | True     |
| fig:autoresearch_closure               | 666x448  | 0.015    | True          | True     |
| fig:autoresearch_integrity             | 440x736  | 0.040    | True          | True     |
| fig:autoresearch_security_matrix       | 472x736  | 0.014    | True          | True     |
| fig:autoresearch_stage_1               | 1120x416 | 0.079    | True          | True     |
| fig:ml_bootstrap_interval              | 1152x448 | 0.009    | True          | True     |
| fig:ml_calibration_reliability         | 1152x832 | 0.015    | True          | True     |
| fig:ml_candidate_rank                  | 1408x560 | 0.053    | True          | True     |
| fig:ml_candidate_score                 | 1376x688 | 0.013    | True          | True     |
| fig:ml_classification_metrics          | 928x832  | 0.092    | True          | True     |
| fig:ml_complexity_accuracy             | 1120x608 | 0.010    | True          | True     |
| fig:ml_confusion_matrix                | 896x768  | 0.045    | True          | True     |
| fig:ml_confusion_pairs                 | 1152x576 | 0.129    | True          | True     |
| fig:ml_generalization_gap              | 1184x864 | 0.059    | True          | True     |
| fig:ml_learning_curves                 | 1216x608 | 0.018    | True          | True     |
| fig:ml_paired_correctness              | 1152x672 | 0.075    | True          | True     |
| fig:ml_per_class_accuracy              | 1152x512 | 0.098    | True          | True     |
| fig:ml_probability_margin_distribution | 1184x864 | 0.024    | True          | True     |
| fig:ml_probability_qualities           | 1134x608 | 0.046    | True          | True     |
| fig:ml_robustness_matrix               | 1280x608 | 0.073    | True          | True     |
| fig:ml_selective_accuracy              | 1088x608 | 0.016    | True          | True     |
| fig:ml_training_dynamics               | 1408x608 | 0.072    | True          | True     |
| fig:mnist_class_balance                | 216x544  | 0.071    | True          | True     |
| fig:mnist_error_examples               | 1280x736 | 0.234    | True          | True     |
| fig:mnist_subset_containments          | 1216x544 | 0.230    | True          | True     |

## Security Readiness And Integrity Evidence

## Security Readiness And Integrity Evidence I

The local security profile reports attestation status passed after checking 80 file record(s), with 0 missing record(s) and 0 checksum mismatch(es). The inventory contains 14 input record(s) and 69 generated-artifact record(s). The integrity-chain figure is fig. 22. These values support local artifact-integrity claims only; they do not claim external signing, production SLSA compliance, or runtime security monitoring.

# Security Readiness And Integrity Evidence I

## Local artifact integrity chain

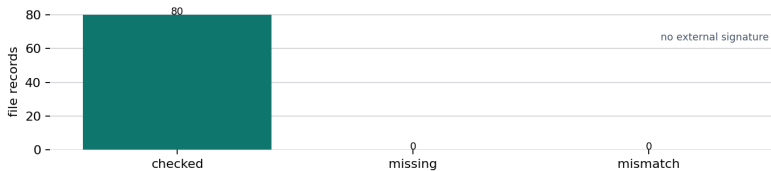


Figure 22: Local integrity chain from output/data/autoresearch\_integrity\_attestation.json; checksums summarize the observed run artifacts and remain unsigned local evidence. Generation method: Local checksum attestation chain with checked, missing, and mismatch counts. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Table 23: Integrity-attestation summary from output/data/autoresearch\_integrity\_attestation.json.

| Integrity field     | Value  |
|---------------------|--------|
| status              | passed |
| algorithm           | sha256 |
| checked files       | 80     |
| missing files       | 0      |
| checksum mismatches | 0      |
| external signature  | False  |

## Manuscript Hydration Provenance

The final run supports 6 manuscript-facing claim(s) and checks 78 required artifact(s). The rendered manuscript uses injected variables from generated data payloads, so the abstract and results track the latest analysis run rather than hard-coded counts. The final readiness status is passed; generated review decisions are recorded as deferred for human review rather than as self-approval.

# Manuscript Hydration Provenance I

Table 24: Variable provenance summary from output/data/manuscript\_variable\_provenance.json.

| Source artifact                                      | Injected variables or fragments |
|------------------------------------------------------|---------------------------------|
| output/data/autoresearch_integrity_attestation.json  | 5                               |
| output/data/autoresearch_loop.json                   | 57                              |
| output/data/autoresearch_phase_ledger.json           | 2                               |
| output/data/autoresearch_schema_manifest.json        | 1                               |
| output/data/autoresearch_security_profile.json       | 6                               |
| output/data/autoresearch_supply_chain_inventory.json | 3                               |
| output/data/autoresearch_threat_model.json           | 4                               |
| output/data/benchmark_scores.json                    | 1                               |
| output/data/figure_quality_report.json               | 3                               |
| output/data/manuscript_variable_provenance.json      | 1                               |
| output/data/ml_bootstrap_intervals.json              | 3                               |
| output/data/ml_calibration_bin_intervals.json        | 1                               |
| output/data/ml_calibration_report.json               | 3                               |
| output/data/ml_candidate_intervals.json              | 1                               |
| output/data/ml_candidate_ledger.json                 | 1                               |
| output/data/ml_candidate_rank_stability.json         | 3                               |
| output/data/ml_candidate_selection_audit.json        | 1                               |
| output/data/ml_class_balance.json                    | 3                               |
| output/data/ml_classification_diagnostics.json       | 5                               |
| output/data/ml_diagnostic_boundary.json              | 1                               |
| output/data/ml_paired_comparison.json                | 3                               |
| output/data/ml_probability_diagnostics.json          | 4                               |
| output/data/ml_robustness_report.json                | 2                               |
| output/data/ml_statistical_summary.json              | 9                               |
| output/data/ml_task_results.json                     | 39                              |
| output/data/ml_training_diagnostics.json             | 7                               |
| output/data/research_object_manifest.json            | 2                               |
| output/data/review_decisions.json                    | 1                               |
| output/figures/figure_registry.json                  | 51                              |
| output/reports/artifact_manifest.json                | 1                               |